Traffic Simulator

User Manual

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1 About

This document describes how to use Traffic Simulator . The software was developed as part of the course System- och programvaruutveckling (DT117G) at Örebro University.

2 Simulation

This section describes the everything related to the simulation mode, from the inner workings of the simulation engine to its visual represation and available user controls.

2.1 Simulation engine

Traffic Simulator uses a simulation model based on the Nagel-Schreckenberger traffic simulation model. The model uses discrete time, and hence moves in textbfsteps or textbfticks.

2.2 Offline Simulation

Traffic Simulator allows for running simulations without any visuals. This is much faster than running simulations "real time". Such a feature can be useful for running very long simulations or multiple simulations quickly. The statistics summary is still shown and can be exported to a CSV file.

2.3 Simulation Visuals

car image	A simulated car
road image	A piece of the road
traffic light image	Traffic light

2.4 Maps

The following maps are implemented*

- **One way street**: A simple one way street. The first map made for this simulator.
- **T Intersection**: A one way street with a connecting one way street in the middle.
- **Highway**: Simple highway with two lanes in each direction. Overtakes are implemented, to allow faster driving vehicles to pass slower driving ones.

- **Autobahn**: A large highway with five lanes in each direction. Supports overtakes in all lanes, and outer lanes has an extra speed up, to ensure the overtaking car does not get stuck in the outer lane.

*more maps will be added in the future, as well as the possibility to create custom maps.

2.5 Live Statistics

1. **Simulated time** from when the simulation started.
2. **Total cars spawned** during the simulation.
3. The number of currently **stationary cars**.
4. The current **number of cars** in the simulation.
5. Current **average speed**.
6. Current **inflow** and **outflow** of cars, respectively. Shown in vehicles per simulated minute.

2.6 Playback Controls

These are the controls available when showing the visuals of a running simulation.

1. **Pause/Resume** the simulation (and the visuals).
2. **Reset** simulation. This will abort the running simulation and start a new one (even if paused) with the same settings.
3. **Stop** the simulation and show statistics summary.
4. **Time scaling** can be adjusted while the simulation is running.

3 Settings Menu

Traffic simulator allows for a number of user settings before starting a simulation. How to use them is described below.

1. **Map selector** Select the map, or scenario, to run a simulation on (see section 2.4 on page 1).
2. **Road length** Set the number of cells in the main road. Only applicable to certain maps. Also note the scaling of vehicles may be very small for large values.

3. **Offline simulation toggle** Allows for running simulations without visuals, and get the statistics summary at the end. Useful for large simulations since it greatly speeds up the simulation time.
4. **Min speed** Set the minimum ideal speed on vehicles.
5. **Max speed** Set the maximum ideal speed on vehicles.
6. **Number of ticks** Only active when running offline simulations. This decides for how long the simulation will run.
7. **Spawn rate** Set the chance of a vehicle spawning each tick.
8. **Time scaling** Set the initial time scaling for the simulation. Not active for offline mode.
9. **Start** simulation Start a new simulation with the current settings.
10. **Help** Open this document in your default PDF viewer.
11. **Exit** program Quit to desktop.

4 Statistics Page

A brief description of the Statistics Page shown after each simulation.

4.1 Statistics Summary

For explanations of 1-6, see 2.5 (page 2).

4.2 Navigation Buttons

1. **Back to Menu** shows the settings menu, where a new simulation can be started.
2. **Export to CSV** lets you export simulation statistics to a CSV file, for further processing or importing to a spreadsheet.
3. **Exit Program** quits to desktop.